

Double-Sided, Thin-Film Microelectrode Array with Hemispheric Electrodes for Subretinal Stimulation

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Components in neural implants, such as the electrode array and stimulator circuit, are often fabricated discretely. This modular fabrication scheme offers flexibility during development but poses difficulties during assembly, as components must be compactly integrated for implantation. It is particularly difficult in cases where the electrode array is required to have a high number of channels, such as in retinal prostheses. This paper presents the development of a parylene C-based, double-sided microelectrode array with 294 hemispheric electrodes for subretinal stimulation. The bonding pads on the bottom side of the double-sided array are connected with electrodes through vias, eliminating the interconnection lines. The array can be integrated with a stimulator circuit through pad-to-pad bonding, resulting in a compact implant. The hemispheric electrodes are fabricated using thermally reflowed photoresist infillings, through which the height and width of the hemispheres can be easily controlled. The long-term stability and biocompatibility of the materials and methods used to fabricate and package the electrodes are demonstrated in *in vitro* and *in vivo* environments over months. Finally, subretinal stimulation by the developed electrodes is successfully demonstrated using *in vitro* retinal patches from mice and monkeys.

1. Introduction

Neural prostheses are devices designed to restore damaged motor, sensory, or cognitive functions. They usually include implantable electrodes that interface with the nervous system for modulating or recording neural activity.^[1] In recent decades, there have been significant advances in neural implants accelerated by microfabrication capabilities adapted from the semiconductor industry.^[2,3] Noticeable trends in the development of neural implants are the increase in the number of electrodes^[4–6] while miniaturizing the device.^[7] For example, monolithic designs using integrated circuit (IC) technologies such as modern complementary metal oxide semiconductors (CMOS) allow multiplexing thousands of neural electrodes with sufficiently high sampling rates.^[8–10]

Despite the advantages of monolithic designs, still many researchers adopt

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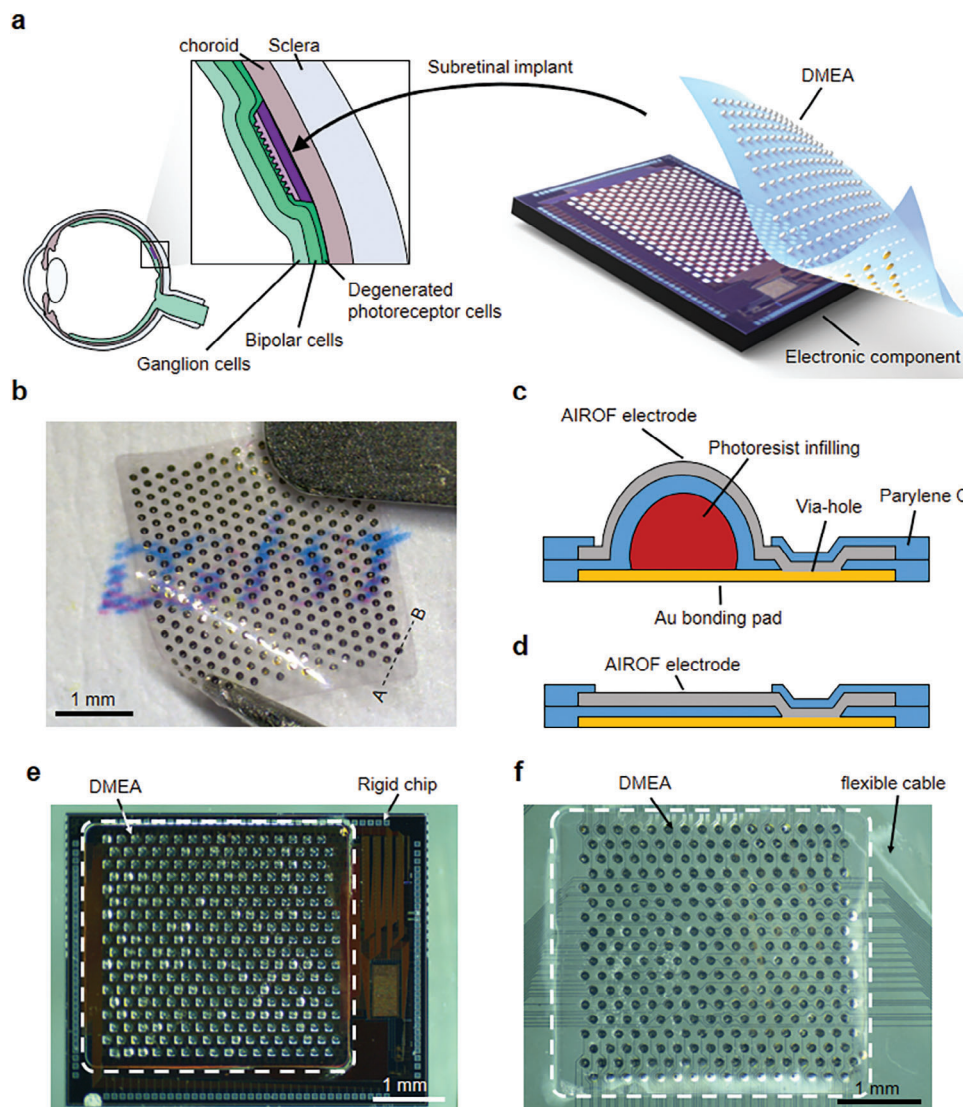


Figure 1. Design and structure of the double-sided, thin-film microelectrode array (DMEA). a) Conceptual illustration of the DMEA bonded to an IC chip for subretinal implantation. b) Photograph of a fabricated DMEA. c, d) Cross-sectional view of c) a hemispheric electrode and d) a disk electrode in the DMEA, cut along with the line A-B in b). e, f) Hemispheric DMEA bonded to e) a rigid stimulator IC chip with 294-pixel photodiode sensors [27] and f) a flexible interconnection cable. The white dashed lines represent the outline of the DMEA.

alternative approaches using modular fabrication schemes.^[11,12] These schemes involve fabricating electrode arrays separately from the signal acquisition or stimulation modules, which are later assembled. Such modular approaches enable fast prototyping with less cost than fabricating the entire implant with a fully integrated IC. In modular approaches, electrodes can be independently tested and revised if needed before the fabrication of IC which places greater demands on circuit and system design. Moreover, some applications demand neural interfaces with specific functionalities or materials that are incompatible with monolithic designs. For example, ECoG and retinal implants often require a flexible electrode array, as the cortex and retina are curved and soft.^[11–14] The flexibility and heat-sensitive polymers for those arrays are not compatible with monolithic IC fabrication.

The assembly involves electrically connecting each electrode with the implantable IC chip or external equipment using connectors and cables. Challenges arise as the number of electrodes increases because the connectors and cables occupy a considerable amount of space and limit the density of electrodes that can be implanted.^[15] Thus, we propose a compact bonding method using a double-sided microelectrode array (DMEA), where the DMEA can be bonded to an IC chip directly in a pad-to-pad manner. The thin film of DMEA has bonding pads on the bottom side and electrodes on the top side. The electrical connections between them are made through vertical vias. The pads of DMEA and IC are bonded using screen-printed, room-temperature curable conductive epoxy paste.

Previous studies have suggested that 3D protruded electrodes for subretinal stimulation, formed by silicon

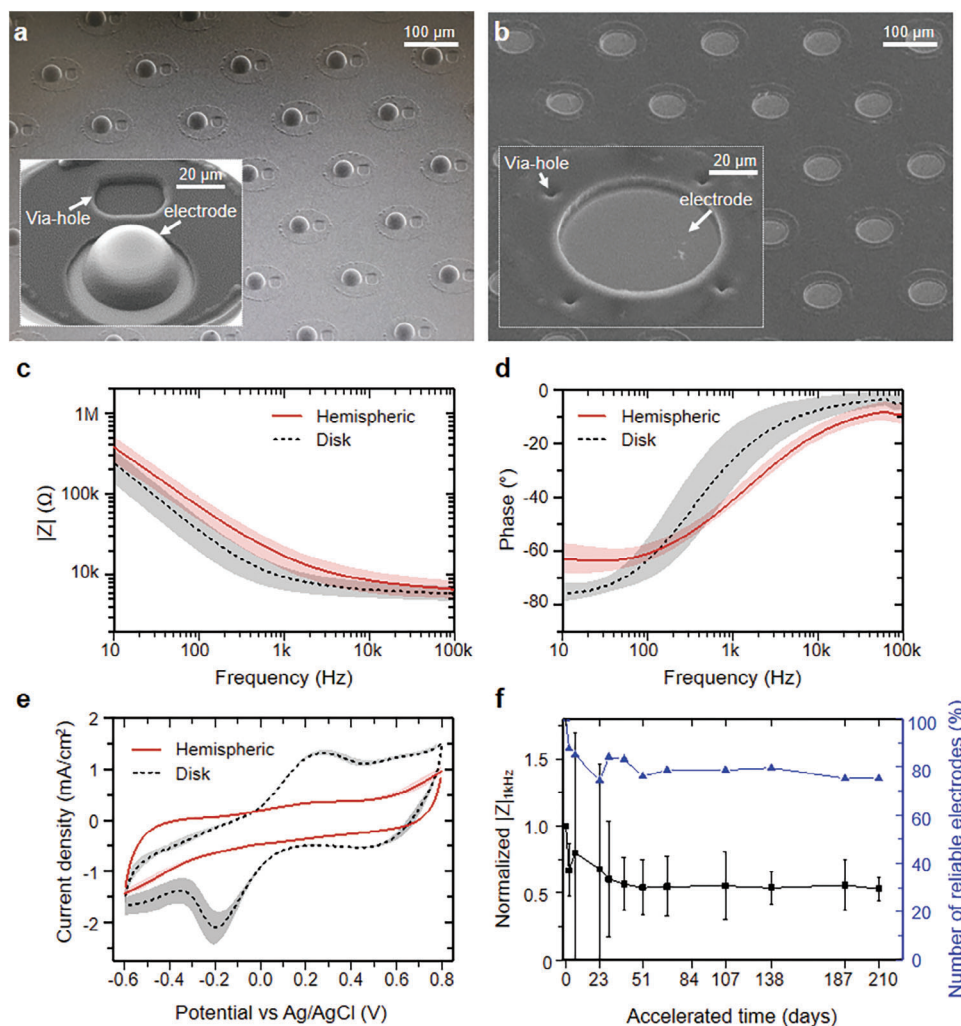


Figure 2. Fabrication results and electrochemical characterization of the electrodes. Scanning electron microscopic images of the DMEA with a) hemispherical and b) disk electrodes. c) Impedance magnitude and d) phase of the hemispherical and disk electrodes measured from EIS. e) CV of the hemispherical and disk electrodes. The shaded regions indicate the standard deviation. f) Long-term monitoring of impedance in an accelerated aging condition for 102 days. The impedance at 1 kHz of reliable electrodes was normalized by the initial impedance. The electrodes with impedance below 100 kΩ were considered reliable.

micromachining,^[16–18] electroplating,^[19,20] or laser milling,^[21] may improve the stimulation efficiency by minimizing the electrode-target distance imposed by residuals in the degenerated photoreceptors layer.^[22] However, pointed corners and sharp edges of the 3D electrodes are known to result in mechanical stresses on the tissue, inducing retinal atrophy,^[23] and concentrated current distributions on the edge, inducing electrode corrosion during stimulation.^[24–26]

In this work, we develop a high-density, double-sided micro-electrode array with 294 hemispherical or disk-shaped electrodes for subretinal stimulation. The DMEA is fabricated based on a thin parylene C film in which electrodes are vertically connected through vias to the bonding pads located on the bottom side of the film. Thus, the DMEA can be bonded to other electronic components, such as an IC chip, through compact pad-to-pad connections. We also develop the bonding method to achieve fine-pitch connections between the DMEA and an electronic com-

ponent using a low-temperature bonding method that is compatible with the polymers used in the device. The electrochemical characteristics, long-term stability, and biocompatibility of the developed electrodes are investigated. The developed electrodes demonstrate effective in vitro subretinal stimulation using retinas from mice and monkeys, confirming that the methods used to fabricate and interconnect the high-density, double-sided, hemispherical electrodes are reliable for subretinal stimulation.

2. Results and Discussion

2.1. Fabrication of Double-Sided Microelectrode Arrays

We developed the DMEA consisting of 294 electrodes for subretinal stimulation. The DMEA was double-sided, with electrodes on the top side of the film and bonding pads on the bottom side, thereby eliminating the interconnection lines between the

electrodes and corresponding bonding pads resulting in a compact implant when assembled with electronic components. **Figure 1a** shows the concept of an assembled subretinal implant, where the DMEA is bonded to a rigid stimulator IC chip. However, the DMEA itself was flexible (**Figure 1b**) and could be bonded to various substrates, regardless of being flexible or rigid. The bonding process included screen-printing conductive epoxy paste on the bonding pads of the DMEA, attaching the DMEA to other electronic components, curing the conductive epoxy paste at room temperature, and underfilling PDMS into the gap between the DMEA and substrate (**Figures S5 and S6**, Supporting Information).

We fabricated two types of DMEAs with hemispheric and disk electrodes (**Figure 1c,d**, respectively), with and without incorporating photoresist infillings. The modularity granted by bonding the DMEA to other electronic components enables various combinations. For instance, a DMEA with hemispheric or disk electrodes can be bonded to a stimulator IC chip with photodiode sensors^[27] (**Figure 1e**). Alternatively, it can be bonded to a flexible cable (**Figure 1f**). The flexible cable in our study was used to interconnect the DMEA with external equipment for electrode characterization and in vitro stimulation. It is noteworthy that the DMEA bonded with the stimulator IC chip as shown in **Figure 1e** was an example of the final implant, which however was not used in experiments of the current study. With such modularity, we could develop electrode arrays independently of the stimulator IC, the development of which requires significant time and cost.

The DMEA was based on a thin parylene C film with a thickness of 6 μm and dimensions of 3.7 mm $\times$ 3.4 mm. The 294 electrodes were hexagonally arranged with a center-to-center spacing of 200 μm , resulting in an electrode density of 29 electrodes/ mm^2 . Each electrode was vertically connected with the corresponding bonding pad on the bottom side of the DMEA through vias. The round shape of the hemispheric electrode was realized through photoresist infilling, which was transformed from a post of cylindrical photoresist through thermal reflow. The thermal reflow method consists of forming photoresist patterns using standard lithography techniques followed by melting them at temperatures above the glass transition temperature of photoresist. The elevated temperature induces a flow of photoresist by reducing viscosity, resulting in a gradual transformation of the photoresist pattern into a spherical form—a technique widely employed in microlens manufacturing.^[28] The reflowed shape can be controlled by various parameters. For example, height by initial photoresist thickness, lateral dimension by lithographic pattern, and roundness by temperature and time.^[29] We chose the round profile which was expected to minimize mechanical stress on the retina^[23] and corrosion of the electrode.^[24–26] The height of the hemispheric electrode, 23 μm , was chosen for closer proximity with target cells, the bipolar cells in the inner nuclear layer (INL). It was reported that the distance between the subretinal implant and the INL in human patients with age-related macular degeneration was $\approx 35 \mu\text{m}$.^[30] Hence, the distance between the dendritic processes of bipolar cells and the subretinal implant was assumed to be closer than 35 μm . As in **Figure 2a**, the hemispherical bump was located in the center of the opening, with a diameter of 50 μm , and coated with activated iridium oxide film (AIROF). The hemispheric bump had a base diameter and height of 36.5 and 23 μm , respectively. A via next to the

opening connected the hemispherical AIROF electrode with an Au bonding pad on the bottom side. The disk electrode, On the other hand, had a flat AIROF active site with a diameter of 70 μm and four vias (**Figure 2b**). The geometrical surface areas of the active sites of the hemispheric and disk electrodes were calculated to be 3380.8 and 3843.5 μm^2 , respectively. The disk electrodes had an AIROF layer activated from a 400 nm-thick Ir layer whereas the AIROF of the hemispheric electrodes was activated from a 200 nm-thick Ir layer.

The resistance through the vias was measured to validate the connection. The fabricated DMEA was placed on a conductive aluminum foil with the bonding pads facing down to establish contact with the foil. Two-point resistance was measured using one probe on the foil and the other on the hemispheric or disk electrode. The measured resistance through one via of the hemispheric electrode was $74.1 \pm 17.9 \Omega$ ($n = 106/106$), and the resistance through four vias of the disk electrode was $69.2 \pm 19.5 \Omega$ ($n = 299/300$, with one over limit), indicating that the electrical connections through the vias were well established.

2.2. Electrochemical Characterization

The hemispheric and disk electrodes coated with AIROF were characterized by electrochemical impedance spectroscopy (EIS) (**Figure 2c,d**) and cyclic voltammetry (CV) (**Figure 2e**). For the characterization, the DMEA was bonded and packaged with an interconnection cable (**Figure S1b**, Supporting Information). The hemispheric electrodes had an impedance at 1 kHz ($|Z|_{1 \text{ kHz}}$) of $17.2 \pm 11.2 \text{ k}\Omega$ ($n = 28$). The disk electrodes had an impedance of $9.4 \pm 6.3 \text{ k}\Omega$ ($n = 32$). The cathodic charge storage capacity (CSC_c) was $10.5 \pm 2.8 \text{ mC cm}^{-2}$ ($n = 5$) for the disk electrodes and $6.3 \pm 1.1 \text{ mC cm}^{-2}$ ($n = 5$) for the hemispheric electrodes, which was sufficient for suprathreshold stimulation of the retina.^[31,32] Notably, the disk electrodes had a thicker AIROF layer, activated from 400 nm-thick Ir, compared with the AIROF of the hemispheric electrodes, activated from 200 nm-thick Ir. The lower impedance and higher CSC_c of the disk electrodes were mainly attributed to the thicker AIROF layer and the larger geometric surface area.^[33] It should be noted that direct comparisons between disk and hemispheric electrodes may not be adequate, as these two electrode types differ not only in shape but also in surface area, AIROF thickness, and the number of vias.

To assess the reliability of the DMEA and its packaging, the impedances at 1 kHz of electrodes were monitored over time in an accelerated aging condition. The DMEAs, bonded to the interconnection cable, were soaked in phosphate-buffered saline (PBS) solution at an elevated temperature of 57 $^\circ\text{C}$ for 102 days, which corresponded to 210 days at 37 $^\circ\text{C}$.^[34,35] Specifically, 30 hemispheric and 66 disk electrodes, which initially had impedances of $28.7 \pm 15.5 \text{ k}\Omega$ and $13.9 \pm 9.9 \text{ k}\Omega$, respectively, were used. During the experimental period, some electrodes exhibited failed connections or abnormally high impedance (**Figure S2**, Supporting Information). The electrodes with an impedance over 100 $\text{k}\Omega$ were considered unreliable because their CSC_c was too high, posing a risk of exceeding the safe potential window of AIROF (-0.6 to 0.8 V vs Ag/AgCl).^[36] The impedance of each reliable electrode was normalized based on its initial value to illustrate the trend in impedance change relative to their

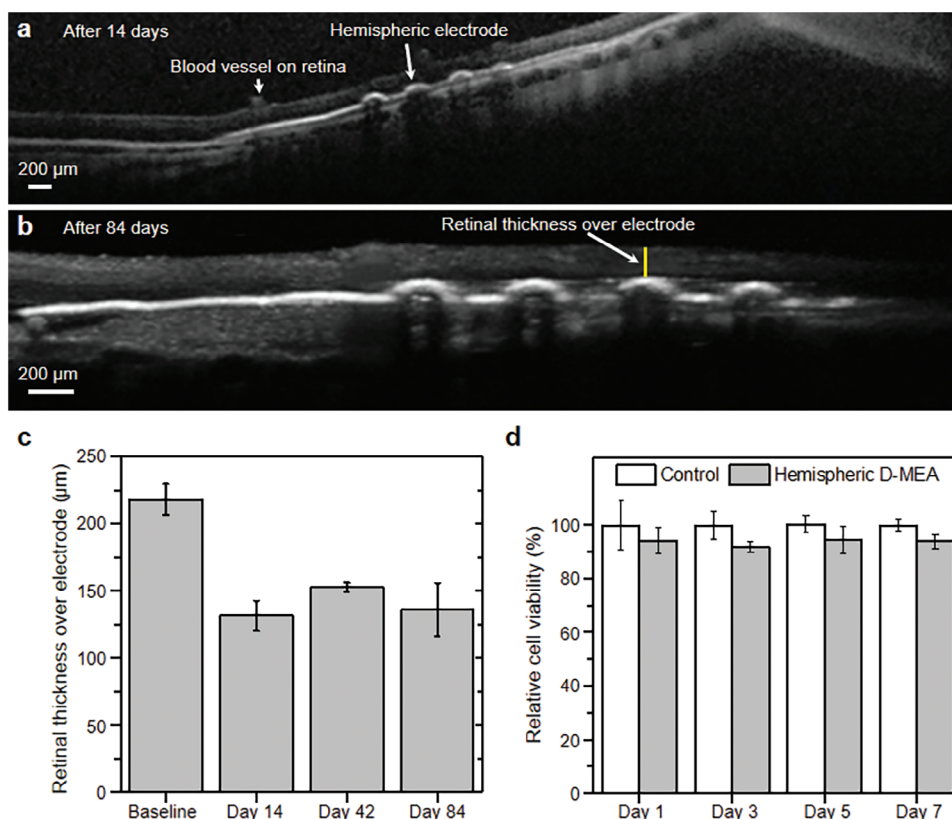


Figure 3. Biocompatibility of the DMEA with hemispheric electrodes. Cross-sectional SD-OCT images of the retina at a) 14 days and b) 84 days after subretinal implantation of hemispheric DMEA in the pig eye in vivo. c) Retinal thickness over the electrode measured before surgery (baseline) and at 14, 42, and 84 days after implantation, which is calculated from the SD-OCT as indicated by the yellow line in b). d) In vitro cell viability evaluation of the hemispheric DMEA using C2C12 cells. The cell viability was normalized with the control group to show the relative cell viability.

initial values (Figure 2f). The impedance and the number of reliable electrodes decreased by 45.8% and 23.9% during the first 50 days (7 weeks). After that, they decreased only by 1.0% and 0.9% at day 210, respectively, compared to day 50. The overall trend in impedance change, decreasing and then stabilizing, is not unusual for electrodes with polymer insulation.^[35,37,38] The decrease in impedance could be caused by water vapor permeated into parylene insulation on interconnection cable, resulting in an increase in insulation capacitance, or by parylene delamination around the electrode, resulting in an increase in the surface area of the electrodes. Scanning electron microscopy (SEM) images taken after accelerated aging (Figure S3, Supporting Information) showed slight delamination of parylene. The delamination revealed some under-etched parylene on the hemispheric electrode, which was difficult to observe before accelerated aging. Additionally, cracks on the AIROF were observed for some electrodes and it could have affected the impedance.

2.3. In Vivo and In Vitro Biocompatibility

DMEAs with hemispheric electrodes were implanted into the subretinal spaces of mini pigs with healthy retinas for 84 days. The used implant was bare DMEAs with a substrate thickness of 6 μm and larger hemispheric electrodes (height of 40 μm

and base diameter of 340 μm), which accentuates the structural change of the retina. Figure 3a,b shows the cross-sectional spectral-domain optical coherence tomography (SD-OCT) images of the retina with hemispheric electrodes subretinally implanted. The hemispheric electrodes and the inner limiting membrane of the retina were clearly observed using SD-OCT, providing nondestructive in vivo visualization. The retinal thickness over the hemispheric electrodes was measured to be $218.1 \pm 11.8 \mu\text{m}$ ($n = 24$) before implantation, decreased to $131.4 \pm 11.2 \mu\text{m}$ ($n = 24$) after 14 days of implantation, and then stabilized afterward (Figure 3c). From a magnified view of SD-OCT, it was confirmed that the inner plexiform layer remained intact over the hemispheric electrodes, and the inner nuclear layer appeared intact or rearranged over the electrodes without tissue damage. The initial decrease in retinal thickness during the first 14 days was mainly attributed to the disappearance of outer retinal layers including photoreceptors, which is considered a normal response of a healthy retina upon subretinal implantation due to blocked metabolic interactions between the retina and retinal pigment epithelium cells.^[39]

In a previous study, it was reported that $\approx 200 \mu\text{m}$ -thick implant with sloped corners and smooth protrusions didn't induce thinning of the inner retinal layer.^[23] The DMEA bonded to the stimulator IC chip in Figure 1e had a total thickness of 199 μm (160 μm for chip, 10 μm for underfiller, 6 μm for DMEA, and

23 μm for hemispheric protrusion). Although we implanted the hemispheric DMEA alone, it did not result in the thinning of the inner nuclear layer, which validates the biocompatibility of the novel, photoresist-infilled, smooth hemispheric electrode.

Additionally, in vitro cytotoxicity of the bare DMEA with hemispheric electrodes was evaluated using a cell viability assay. The cell viability was observed to be $94.0 \pm 4.8\%$ after 24 h and $93.8 \pm 2.6\%$ after 7 days (Figure 3d). According to the ISO protocol (ISO10993-5), a material is considered noncytotoxic if the relative cell viability percentage is $>70\%$ of the control group after exposure for 24 h. Throughout the experimental period of 7 days, the normalized viability percentages remained $>90\%$, demonstrating that the electrode array, which incorporates photoresist infillings, was effectively encapsulated by parylene C and did not exhibit cytotoxic effects.

2.4. In Vitro Retinal Stimulation

The efficacy of the fabricated electrodes for in vitro subretinal stimulation was evaluated through simultaneous stimulation and recording of retinal patches of wild-type (WT) mice and retinal degenerated (RD) monkeys (Figure 4a). Here, the DMEAs bonded to flexible cables were used. Figure 4b shows an example of raw signal obtained from a mouse retina upon subretinal stimulation. The bursts of spikes were identified mainly ≈ 10 –40 ms after stimulation onset, indicating the indirect activation of retinal ganglion cells (RGCs) through synaptic relays.^[40] The responsive RGC spikes of the mouse retinas upon stimulation using hemispheric and disk electrodes were identified (Figure 4c,d). The threshold was calculated as a current value that induces 50% of the maximum RGC response. The threshold of WT mice was 22.03 and 17.12 μA for a hemispheric electrode and a disk electrode, respectively, which is consistent with reported threshold charge densities for subretinal stimulation of WT mice.^[41] As the stimulation amplitude increased, both the number of spikes and the responsive area increased up to 30 μA , beyond which they gradually saturated (Figure 4e,f). In Figure 4g, the hemispheric electrode evoked higher responses at closer proximity. On the other hand, the disk electrode induced more deviated responses with peak responses at $\approx 300 \mu\text{m}$ distance from the stimulating electrode, as shown in Figure 4h. It was clear that the responses of the hemispheric electrode were more focused around the stimulating electrode, compared to the disk electrode. The protruded shape and smaller footprint of the hemispheric electrode may have contributed to a more spatially confined response.^[16] However, It should be noted that the preparation of the retinal patch also had a strong impact on the threshold and spatial response. The condition of the retinal patch, the preparation technique, and the contact between the electrode and tissue affect the signal quality.

To further assess the efficacy of the fabricated electrodes, subretinal stimulation was conducted using retinal patches from RD-monkeys, which was considered to provide a closer approximation to human patients with retinal degeneration. Unlike the WT-mouse retina, the RD-monkey retina showed multiple rhythmic bursts of spikes upon stimulation (Figure 5a,b), which is the characteristic behavior of the photoreceptor degenerated retina model upon electrical stimulation.^[42–44] The RGC response of the RD-

monkey could be well modulated with increasing pulse amplitude using both hemispheric and disk electrodes (Figure 5c). The threshold of RD-monkey was 25.01 μA for the disk electrode and 23.93 μA for the hemispheric electrode. To our knowledge, there is no reported threshold for RD-monkey in subretinal stimulation. However, the obtained thresholds were similar to the reported thresholds of RD-monkey in epiretinal stimulation.^[44,45]

3. Conclusion

To address the challenges associated with the interconnection of compact neural implants including high-density electrode arrays, we developed double-sided microelectrode arrays based on parylene C thin films. For hemispheric electrodes, the electrode shape can be easily modified by controlling the height and width of the photoresist infillings that are thermally reflowed and sealed using parylene C. To our knowledge, this study is the first to suggest the novel concept of the double-sided electrode array in the form of a polymer thin film with 3D electrode structures produced by permanent photoresist infilling. The compact, high-density pad-to-pad bonding was achieved by incorporating vias in the double-sided electrode array. A low-temperature bonding method using screen-printed conductive epoxy was employed to connect the thin-film-based high-density electrodes to other electronic components. The long-term stability of the DMEA and bonding materials was investigated in an accelerated aging condition. The biocompatibility of the structure and materials used in the developed electrodes was proven in in vivo and in vitro. Finally, we demonstrated subretinal stimulation by the developed double-sided electrodes in in vitro experiments using retinas from mice and monkeys. The simple fabrication of the double-sided, high-density, hemispheric electrode array is expected to be suitable for batch processing while providing process flexibility during development and effective subretinal stimulation during application.

4. Experimental Section

Fabrication of the DMEA: The fabrication of the DMEA with hemispheric electrodes is explained first. The overall process flow of the DMEA with disk electrodes was the same with the hemispheric electrodes, except for the photoresist infilling steps (Figure S4 Supporting Information, steps 2 and 3) and some minor differences to be explained later. First, a 100 nm-thick Ti sacrificial layer was sputtered on a silicon wafer. A 200 nm-thick Au layer was sputtered as a bottom metal layer and patterned using a wet etching process with a photoresist (AZ5214E, MicroChemicals GmbH, Germany) (Figure S4, Supporting Information, step 1). This step defined the 294 bonding pads with a diameter of 100 μm . The Au surface was roughened with 0.05 mol L⁻¹ iodine solution for 20 s to enhance the adhesion with the subsequent parylene C layer.^[46] For hemispheric electrodes, a 25 μm -thick positive photoresist (AZ 40XT-11D, MicroChemicals GmbH, Germany) was spin-coated and patterned by photolithography to form a 30 μm -diameter cylindrical column on each Au bonding pad (Figure S4, Supporting Information, step 2). The wafer was placed in a convection oven at 125 $^{\circ}\text{C}$ for 1 min to induce thermal reflow, during which the cylindrical structures of the photoresist converted to hemispheres. These hemispheric structures remained as permanent infillings (Figure S4, Supporting Information, step 3). A 3 μm -thick parylene C layer was deposited by chemical vapor deposition (CVD) (NRPC-500, Nuritech, Korea) to conformally cover the surfaces including the hemispheric bumps. Next, the holes in the parylene C layer were etched to expose a portion of the bottom metal layer by O₂ reactive ion etching (RIE) (VITA, Femtosience, Korea),

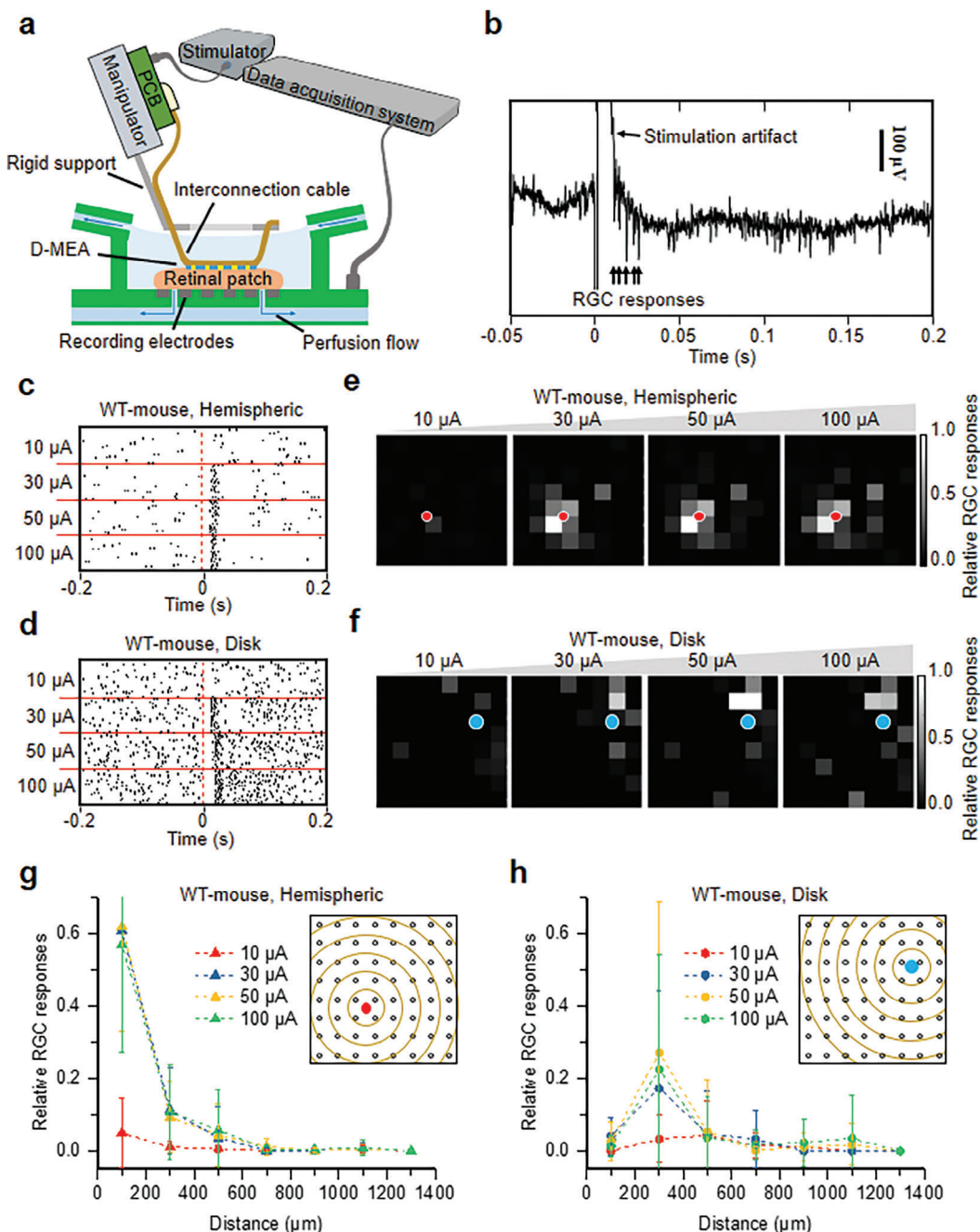


Figure 4. In vitro subretinal stimulation using retinas from WT-mouse. a) Schematic illustration of the in vitro subretinal stimulation and simultaneous recording. b) Representative recorded waveform showing electrically evoked RGC spikes upon stimulus with a pulse amplitude of 100 μ A and duration of 500 μ s per phase applied at 0 s. c,d) Raster plots of RGC spikes measured from a representative recording electrode upon subretinal stimulation using a c) hemispheric and d) disk electrode. e,f) Spatial distribution of RGC responses upon stimulation using e) hemispheric and f) disk electrode. g,h) Relative RGC responses as a function of distance from the stimulating electrode upon stimulation using g) hemispheric and h) disk electrode. 64 recording electrodes were categorized into 7 groups according to the distance from the stimulation electrode with 200 μ m-interval. The responses within each group were averaged.

using a patterned photoresist layer (AZ 40XT-11D) as a mask. These holes defined a rectangular via with an area of 30 μ m $\times$ 20 μ m, located on each bonding pad (Figure S4, Supporting Information, step 4). Top metal layers of Cr and Ir, with thicknesses of 50 and 200 nm, respectively, were sputter deposited and patterned by a lift-off process using AZ 40XT-11D, defining

the stimulating electrodes on the hemispheric bumps (Figure S4, Supporting Information, step 5). A second layer of 3 μ m-thick parylene C was deposited by CVD and patterned by O₂ RIE, using a patterned photoresist AZ 40XT-11D layer as the mask, to define the openings of the stimulating electrodes and outlines of each DMEA (Figure S4, Supporting Information,

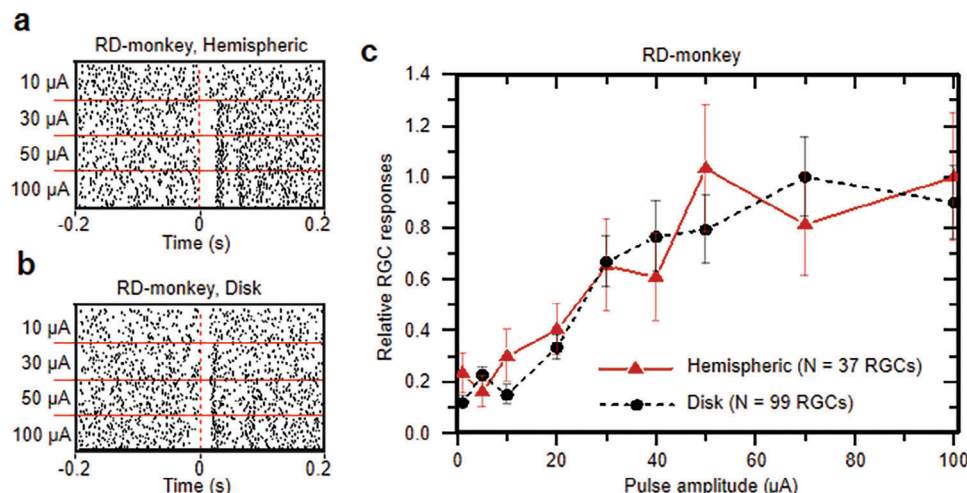


Figure 5. In vitro subretinal stimulation using retinas from RD-monkey. a,b) Raster plots of RGC spikes measured with a representative recording electrode upon subretinal stimulation using a) hemispheric and b) disk electrode. c) Relative RGC responses of RD-monkey retinas with increasing pulse amplitude.

step 6). The DMEAs were released from the wafer by Ti wet etching (Figure S4, Supporting Information, step 7). The minor differences in the fabrication of the disk DMEA were that it had four vias with diameters of 10 μm and a thicker Ir layer with 400 nm.

Bonding and Packaging of the DMEA: First, a stencil mask was prepared. The stencil mask was made of SU-8 (SU-8 3010, Kayaku Advanced Materials, Japan) and had a thickness of 15 μm and 294 apertures with a diameter of 60 μm (Figure S5, Supporting Information, step 1). The apertures of the stencil and the bonding pads of DMEA were aligned (Figure S5, Supporting Information, step 2). Two components of conductive epoxy adhesive paste (Duralco 125-2, Contronics, USA) were thoroughly mixed at a ratio of 1:1. The mixed paste was put on the stencil and, using a squeegee, spread across the 294 apertures (Figure S5, Supporting Information, step 3). The stencil was then removed, leaving the 294 posts of conductive epoxy paste on the bonding pads (Figure S5, Supporting Information, step 4). At this step, the minimum diameter of conductive epoxy paste was limited. The stencil was successfully removed with apertures of 60 μm-diameter, leaving behind the conductive epoxy paste on the bonding pads. However, when the stencil was tried to be removed to print a 25 μm-diameter paste, the paste was not properly printed. Next, the DMEA was mounted on a micromanipulator, and aligned with an electronic component having corresponding bonding pads, such as interconnection cable or stimulator IC chip as shown in Figures S1 and S5 (Supporting Information, step 5). After the alignment, the DMEA was slowly lowered toward the electronic component. The final bonding pressure was 0.15 kg, which compressed the 15 μm-tall posts of conductive epoxy paste to ≈10 μm. Then, the conductive epoxy paste was cured at room temperature for 24 h, providing mechanical and electrical connections between the DMEA and the electronic component (Figure S5, Supporting Information, step 6). PDMS (Sylgard 184, Dow Corning, USA), mixed at a ratio of 10:1 (base: curing agent), was injected into the gap between the DMEA and the electronic component using a syringe, for mechanical stability and electrical insulation of the conductive epoxy. PDMS was cured at room temperature for 24 h (Figure S5, Supporting Information, step 7). A final layer of 3 μm-thick parylene C was applied by CVD for packaging the entire assembled device except for the electrode sites (Figure S5, Supporting Information, step 8). During the parylene C deposition, a rectangular sheet of polyethylene glycol (PEG) was placed on the electrode array to mask 294 electrodes. After the parylene C deposition, the PEG sheet was removed with tweezers, and residual PEG was dissolved in warm water. Finally, Ir on the electrodes was converted into an AIROF by sweeping the voltage potential between +1 and −1 V (vs Ag|AgCl reference electrode), for 100 cycles with a rate

of 100 mVs^{−1}. The fabrication methods for the interconnection cable [47] and the stimulator IC chip [27] are described in previous studies.

Electrochemical Characterization: The interconnection cable bonded with a DMEA and packaged by parylene-C was connected to a potentiostat (Reference 600+, Gamry Instruments, USA). An interconnection cable could connect 143 of the 294 electrodes on the array (Figure S1b, Supporting Information). EIS and CV were performed using a three-electrodes configuration. A disk or hemispheric electrode was used as the working electrode, while an Ag|AgCl electrode and a Pt wire served as the reference and counter electrodes, respectively. PBS (LB 004-02, Welgene, Korea) was used as the electrolyte. For EIS measurements, the impedance magnitude and phase were measured while a sinusoidal input voltage of 10 mV_{rms} was applied at frequencies ranging from 100 kHz to 10 Hz. For CV, the current density was measured while the working electrode potential was ramped between +0.80 and −0.60 V using a triangular waveform with a rate of 100 mVs^{−1}.

Impedance Monitoring in Accelerated Aging Condition: To estimate the long-term stability of the bonded DMEA, the impedance at 1 kHz of 96 electrodes (66 disk and 30 hemispheric electrodes) was monitored under an accelerated aging condition for 102 days. The DMEA, bonded and packaged with an interconnection cable, was soaked in PBS in a glass container and stored in an incubator at 57 °C with a relative humidity of > 80%. The elevated temperature enables an accelerated simulation of aging, which is frequently used for the assessment of polymeric materials associated with medical devices.[34] The acceleration constant of parylene C calculated from the Arrhenius equation was 2.0585.[35] The electrochemical impedance of the electrodes was measured over time for 102 days at 57 °C, which corresponded to 210 days at 37 °C. To prevent evaporation, the glass container was sealed using an air-tight lid with connectors on it, such that zero-insertion-force connectors from the potentiostat could be connected repeatedly without opening the lid. As the majority of channels had impedances of less than 100 kΩ with a gradual decrease over time (Figure S2, Supporting Information), the channels exceeding 100 kΩ were considered unreliable. The impedance of each reliable electrode was normalized using its initial value to show the fluctuation of impedances relative to their initial values.

In Vivo Biocompatibility Evaluation: To assess the biocompatibility of the developed electrodes, three bare DMEAs with hemispheric electrodes were implanted into the subretinal spaces of mini pigs for 84 days. Three female mini pigs (APURES Co. Ltd., Korea) were used with normal retinas (details for animal protocols in the Supporting Information). Retinal changes were investigated using SD-OCT images, obtained before surgery

and at 14, 42, and 84 days after implantation. The DMEAs used in this experiment were 4 by 4 arrays with larger sizes of hemispheric electrodes with a height of 40 μm , base diameter of 340 μm , and pitch of 555 μm . SD-OCT images of the fundus were obtained using the Spectralis OCT system (Heidelberg Engineering GmbH, Germany). Vertical and horizontal line scans, as well as raster scans (37 B-scans over an area of 16.5 mm $\times$ 16.5 mm in a 55-degree image), were performed at high resolution (1536 A-scans per B-scan, lateral resolution = 10 $\mu\text{m}/\text{pixel}$ in a 55-degree image). Up to 100 images were averaged in automatic real-time mode to obtain a high-quality mean image. The total retinal layer thickness before surgery was measured along a line perpendicular to the retinal layers in cross-sectional images. The total retinal thicknesses at 14, 42, and 84 days post-implantation were measured over eight hemispheric electrodes (four points in the central area and one point at each of four marginal areas of the array). The total retinal thickness over the electrode was defined as the distance between the tip of each electrode and the inner margin of the internal limiting membrane (Figure 3b).

In Vitro Cytotoxicity Evaluation: Three bare hemispheric DMEAs were used for in vitro cytotoxicity evaluation. The arrays were placed inside three wells of a polystyrene well plate (00 307 21110, Eppendorf, Germany), with the hemispheric electrode side facing up. C2C12 cells were then seeded onto the surface of the electrode arrays at a density of 20 000 cells per well. In another three wells, serving as the control group, C2C12 cells were seeded onto the bare polystyrene surface of the well plate at the same cell density. The cells were cultured for a week using DMEM (LM 001–05, Welgene, Korea) with 10% fetal bovine serum and 1% antibiotic-antimycotic. Cell viability was measured at culture days 1, 3, 5, and 7 using a CCK-8 kit (CK04, Dojindo, Japan), according to the procedure recommended by the manufacturer.^[48] The viability percentages were normalized to the control group.

In Vitro Subretinal Stimulation: Retinal patches were isolated from two 10-week-old wild-type mice (C57BL/6J strain) and two adult male cynomolgus monkeys (*Macaca fascicularis*) with outer retinal degeneration induced by N-methyl-N-nitrosourea^[49] (Figure S7a,b, Supporting Information). The retinal patch was attached to the recording electrode array (60pMEA200/30iR-Ti, Multichannel Systems GmbH, Germany) with the ganglion cell layer facing down contacting recording electrodes. After the attachment, the retinal tissue was allowed to stabilize for 20 min before recording. The recording electrode array consisted of 59 titanium nitride electrodes, each with a diameter of 30 μm and an inter-electrode distance of 200 μm (Figure S7c,d, Supporting Information). The extracellular potentials of RGCs were recorded from the recording electrodes and processed using a data acquisition system (MEA2100 system, Multichannel Systems GmbH, Germany). For more detailed information regarding the MEA system, refer to the previous study.^[17] During the recording process, continuous perfusion of oxygenated fresh ACSF was maintained on the retinal tissue.

The fabricated electrodes, bonded to an interconnection cable, were mounted on a micromanipulator and positioned on the retinal patch in a manner such that they faced the photoreceptor layer of the retina (Figure 4a). The retinal patch, sandwiched between the DMEA and recording electrodes, was very susceptible to pressure. The DMEA bonded to the cable was flexible and suspended on the rigid support.^[42] The flexibility of the DMEA and cable was helpful to minimize the pressure applied to the retinal patch. For electrical stimulation, a stimulator (STG 2004, Multichannel Systems GmbH, Germany) was employed. The interconnection cable was connected to the stimulator through a zero-insertion-force connector (5 025 983 393, Molex, United States) that is surface-mounted on a printed circuit board (PCB). The current pulse train was applied at a frequency of 1 Hz with cathodic phase-first biphasic symmetric pulses. To assess the effectiveness of the fabricated electrodes in eliciting RGC spikes, the current amplitude was varied from 1 to 100 μA . It is important to note that the subretinal stimulation approach primarily targeted the activation of bipolar cells located in the inner nuclear layer, whereby the electrically evoked responses of bipolar cells are transmitted to RGCs via synaptic relays, called bipolar cell-mediated RGC response.^[40] A pulse duration of 500 μs per phase was adopted, which has been widely established as an optimal stimulus parameter for specifically activating bipolar cells.^[50] As

our main focus did not concern directly-evoked RGC responses, any RGC spikes occurring within 10 ms after the onset of the electrical stimulation were excluded.

Signal Processing and Analysis: The raw signal trace was processed using a high-pass filter with a cut-off frequency of 100 Hz to eliminate low-frequency components such as 60 Hz noise. Then, the filtered signal underwent spike sorting using Offline Sorter (Plexon Inc., USA). Principal component analysis was used to separate multiunit activities with different spike waveforms into individual cell units. Once the RGC spikes were isolated, their timestamps were extracted and further analyzed using both NeuroExplorer (Nex Technologies, USA) and custom-made MATLAB scripts (MathWorks, USA). To quantify the electrically evoked RGC responses, the average spike number before stimulation was determined by counting the spikes observed during a 100 ms period before the onset of stimulation across 20 trials. Next, the average spike number after stimulation was calculated within a 100 ms period following the stimulation across 20 trials. Then, the average spike number before stimulation was subtracted from the average spike number after stimulation, to represent the electrically evoked RGC response. To obtain the relative RGC response, the evoked spike number for each pulse amplitude was divided by the maximum evoked spike number. During the in vitro stimulation, 69 individual RGCs could be identified from the recorded signals of a WT-mouse retinal patch upon stimulation using a disk electrode, 33 RGCs from another WT-mouse retinal patch using a hemispheric electrode, 99 RGCs from an RD-mouse retinal patch using a disk electrode, and 37 RGCs from another RD-mouse retinal patch using a hemispheric electrode.

Ethics Approval Statement: All procedures involving animals in this study followed the guidelines of the Association for Research in Vision and Ophthalmology Statement for the Use of Animals in Ophthalmic and Vision Research. The protocols to use mice, monkeys, and mini pigs were approved by the Institutional Animal Care and Use Committees (IACUCs) of Chungbuk National University (CBNUA-1708-22-01), Osong Medical Innovation Foundation (KBIO-IACUC-2020-054-4) and Korea University (KOREA-2018-0002-C3), respectively.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

double-sided microelectrode array, hemispheric electrode, retinal prostheses, subretinal stimulation

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